[redacted]

VBF-T7R3-CALC-01 Case t7_r3 | 2026-08-26

Prepared: _____________ Reviewed: _____________ Approved: _____________

Virtual design package - simulation caliber (no physical build)

VBF-T7R3-CALC-01 Case t7_r3 | 2026-08-26 - input_parameters

Parameter	Value (final design)	Unit	Base (Chen2020)	Changed by	Source
Nominal cell capacity	6.28	Ah	5	yes (honest re-rating)	r7_v19_final_params.json
Positive electrode thickness	7.56e-05	m	7.56e-05	no	Chen2020 set
Negative electrode thickness	0.00011	m	8.52e-05	yes	r7_v19_final_params.json
Separator thickness	1.2e-05	m	1.2e-05	no	Chen2020 set
Positive particle radius	2e-06	m	5.22e-06	yes	r7_v19_final_params.json
Negative particle radius	2e-06	m	5.86e-06	yes	r7_v19_final_params.json
Positive porosity	0.335	-	0.335	no	Chen2020 set
Negative porosity	0.25	-	0.25	no	Chen2020 set
Positive active volume fraction	0.665	-	0.665	no	Chen2020 set
Cation transference number	0.6	-	0.2594	yes	r7_v19_final_params.json
Electrolyte conductivity	1.5	S/m	Nyman2008 fn	yes	r7_v19_final_params.json
Electrolyte diffusivity	1e-09	m2/s	Nyman2008 fn	yes	r7_v19_final_params.json
Upper / lower voltage cut-off	4.2 / 2.5	V	same	no	Chen2020 set
Electrode height x width	0.065 x 1.58	m	same	no	Chen2020 set
Ambient temp (charge/aging)	318.15	K	-	protocol	aging_1C_100cyc_45C / 4C_charge_45C

VBF-T7R3-CALC-01 Case t7_r3 | 2026-08-26 - capacity_energy

Quantity	Formula	Value	Unit	Source
1C discharge capacity (SPMe)	protocol 1C_discharge, CC 6.28 A	6.2569514025042058	Ah	r7_v19_final_1c.json:capacity_ah
1C discharge capacity (DFN)	protocol 1C_discharge DFN	6.28	Ah	r6_v19_1c_dfn.json
Rated energy	integral V x I_1C dt / 3600, I_1C = 6.28 A	26.498098322496203	Wh	energy.json:energy_wh
Midpoint voltage	V at 50% of discharge time	4.047268213948273	V	energy.json
DCR	(V(t0) - V(t=10%)) / I_1C	0.0001200748574169952	ohm	energy.json:dcr_ohm
Aging SEI end (100cyc@45C)	SEI ec-reaction-limited model	512.9188471770843	nm	r7_v19_final_aging45c.json
4C anode surface potential min	DFN + plating, 25.12 A charge	0.01709696400542111	V	r7_v19_final_4c45c.json:anode_potential_v
4C T_max	lumped thermal	363.2303675149965	K	r7_v19_final_4c45c.json:T_max_K

VBF-T7R3-CALC-01 Case t7_r3 | 2026-08-26 - energy_density

Quantity	Formula	Value	Unit	Source
Contract mass (electrolyte excluded)	sum layer thickness x (1-porosity) x density x area	46.61976206160001	g	energy.json:mass_kg
Energy density (contract caliber)	E_Wh / mass_kg	482.4345348490637	Wh/kg	energy.json:energy_density_wh_kg
Threshold (task text)		327.18	Wh/kg	log.jsonl entry 0 stage2
Electrolyte mass add-on	pore volume 6.004e-6 m3 x 1200 kg/m3 (lit.)	7.21	g	derived (literature density 1.2 g/cm3)
Energy density incl. electrolyte	22.49098 / (0.0466198 + 0.00721)	417.8165853293534	Wh/kg	derived
Volumetric energy density	E_Wh / (stack thickness 225.6e-6 x area m3)	970.7831005563035	Wh/L	energy.json
Power density (peak proxy)	Voc^2 / (4 x DCR) / mass	748494.3392104406	W/kg	energy.json:power_density_w_kg

VBF-T7R3-CALC-01 Case t7_r3 | 2026-08-26 - np_and_mass

Quantity	Formula	Value	Unit	Source
Positive areal Li capacity	63104 x 0.665 x 75.6e-6	3.1725	mol/m2	Chen2020 max conc + geometry
Negative areal Li capacity	33133 x 0.75 x 110e-6	2.7335	mol/m2	Chen2020 max conc + geometry
N/P ratio	neg/pos areal capacity	0.8616233254531127	-	mechanical derivation
Positive electrode mass	164.0 g/m2 x 0.1027 m2	16.8421620276	g	energy.json layer_kg_m2 x area
Negative electrode mass	136.7 g/m2 x 0.1027 m2	14.03934675	g	energy.json layer_kg_m2 x area
Al collector mass	43.2 g/m2 x 0.1027 m2	4.436640000000001	g	energy.json layer_kg_m2
Cu collector mass	107.5 g/m2 x 0.1027 m2	11.042304	g	energy.json layer_kg_m2
Separator mass	2.52 g/m2 x 0.1027 m2	0.259309284	g	energy.json layer_kg_m2
Total contract mass	sum of layers	46.6197620616	g	energy.json:mass_kg x1000

VBF-T7R3-CALC-01 Case t7_r3 | 2026-08-26 - process_parameters

Process parameter	Formula	Value	Unit	Source
Positive areal density	thickness x (1-porosity) x density	164	g/m2	energy.json layer_kg_m2
Negative areal density	thickness x (1-porosity) x density	136.7	g/m2	energy.json layer_kg_m2
Positive compaction density	3262 x (1-0.335) / 1000	2.16923	g/cm3	Chen2020 density x (1-porosity) / 1000
Negative compaction density	1657 x (1-0.25) / 1000	1.24275	g/cm3	Chen2020 density x (1-porosity) / 1000
Electrolyte fill amount	pore volume x 1200 kg/m3	7.21	g/cell	6.004 mL x 1.2 g/cm3 (literature)
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	-	-	design-recommended value; production requires tu